

SOFTWARE ENGINEERING

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement 60 — Podcast Guest Scheduling & Outline Builder

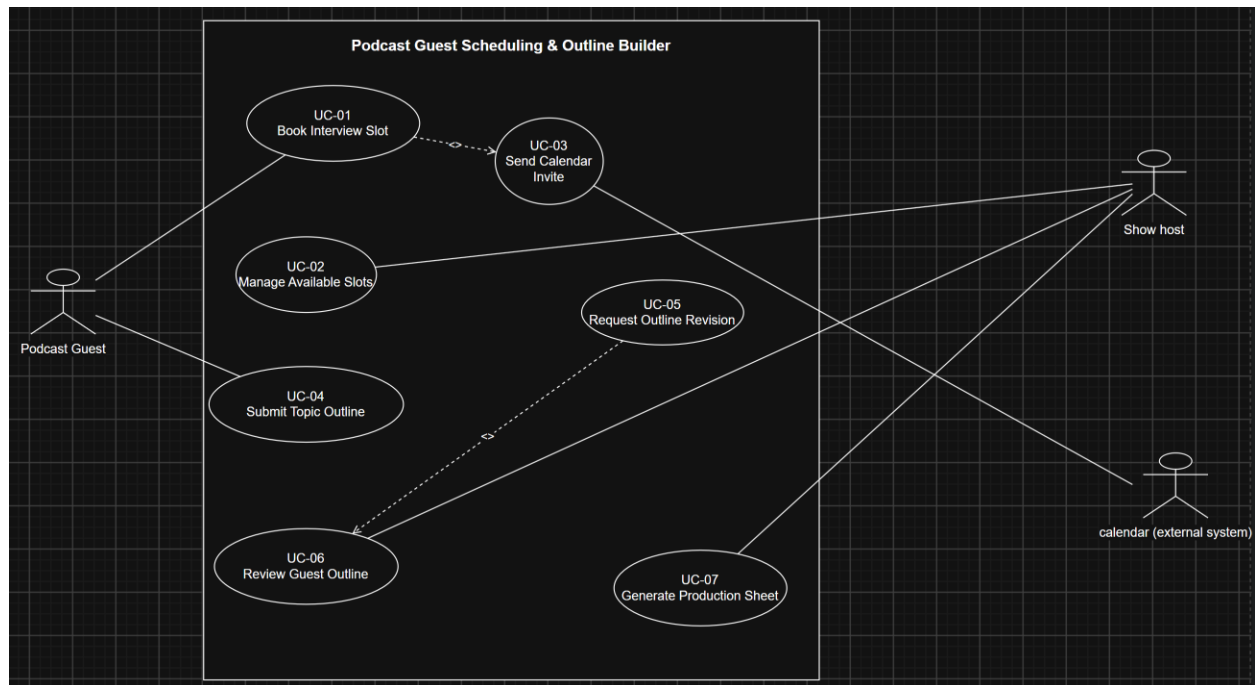
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Section 5-G

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow guests to select interview time slots and submit structured bullet points for discussion topics with biography links.	High	Pass: Slot booked, calendar invite sent, and topic outline attached to episode board. Fail: Booking permitted on blocked dates.	Core booking function given in the problem statement; enables the scheduling workflow the rest of the system depends on.
FR-002	Functional	The system shall allow the show host to define and update available interview time slots, including blocking out specific dates.	High	Pass: A date marked blocked by the host no longer appears as selectable in the guest slot picker. Fail: A guest can book a blocked date.	Prevents double-booking and scheduling conflicts with the host's existing commitments.
FR-003	Functional	The system shall automatically generate and send a calendar invite containing the meeting link to both guest and host within 2 minutes of a slot being booked.	High	Pass: Both parties receive an invite with correct date, time, and link within 2 minutes of booking. Fail: No invite is sent, or invite details are incorrect.	Reduces missed interviews and confirms the booking is recognised by both parties.
FR-004	Functional	The system shall allow the show host to review a guest's topic outline after it has been attached to the episode board, and to flag specific items as needing revision, triggering an optional revision request back to the guest.	Medium	Pass: The host can view the attached outline on the episode board and mark items as 'needs revision', which sends a revision request to the guest. Fail: The host cannot flag items, or the original attached outline is lost when a revision is requested.	Gives the host editorial input on episode content quality without delaying the immediate attachment guaranteed by FR-001.
FR-005	Functional	The system shall compile all approved guest topic outlines for an episode into a timestamped run-of-show production sheet.	High	Pass: The production sheet lists every approved topic in sequence with an assigned timestamp and the guest's biography link. Fail: A topic,	Delivers the core production artefact the host needs to run the recording session.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
				timestamp, or bio link is missing from the sheet.	
NFR-001	Nonfunctional (Performance)	The show outline generator must compile submitted guest topics into a formatted PDF production sheet in under 1 second.	High	Pass: Benchmarking tests confirm PDF generation latency is consistently under 1 second under simulated peak load. Fail: Generation latency exceeds 1 second.	Hosts often generate the production sheet minutes before recording, so fast turnaround is critical to a smooth session.
NFR-002	Nonfunctional	The guest-facing booking interface shall be fully usable on both desktop and mobile browsers without requiring a dedicated app install, and shall load within 3 seconds on a standard broadband connection.	Medium	Pass: The booking form renders correctly and is fully functional on Chrome desktop and Chrome mobile, loading in under 3 seconds. Fail: The layout breaks on either platform or load time exceeds 3 seconds.	Guests frequently book slots on the go; a frictionless, app-free experience increases booking completion rates.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case: UC-01 — Book Interview Slot

Primary Actor: Podcast Guest

Supporting Actors: Show Host, Calendar Service

Preconditions

- The guest has received a valid booking link from the show host.
- The show host has published at least one open (unblocked) interview time slot.

Postconditions

- Success: The selected slot is reserved for the guest, marked unavailable to other guests, a calendar invite (UC-03) has been sent to both parties, and the guest is directed to the topic outline submission screen (UC-04).
- Failure: No slot is reserved and the guest's calendar/availability state is unchanged.

Main Success Scenario

1. The guest opens the booking link shared by the show host.
2. The system displays the list of currently available interview time slots, drawn from the host's published availability (UC-02).
3. The guest selects a desired time slot.
4. The system validates that the selected slot is still open.
5. The system reserves the slot for the guest and marks it unavailable to all other guests.
6. The system sends a calendar invite with the meeting link to both the guest and the host (include: UC-03 — Send Calendar Invite).
7. The system displays a booking confirmation and prompts the guest to proceed to topic outline submission.
8. The guest proceeds to submit their topic outline (UC-04 — Submit Topic Outline).
9. The use case ends successfully.

Alternate Flow — 4a: Slot No Longer Available

- 4a.1 A concurrent booking has already reserved the slot the guest selected. The system detects the conflict during validation (Step 4).
- 4a.2 The system displays a "slot no longer available" message and refreshes the list of open slots.
- 4a.3 The guest selects a different available slot; the flow resumes at Main Success Scenario Step 3.